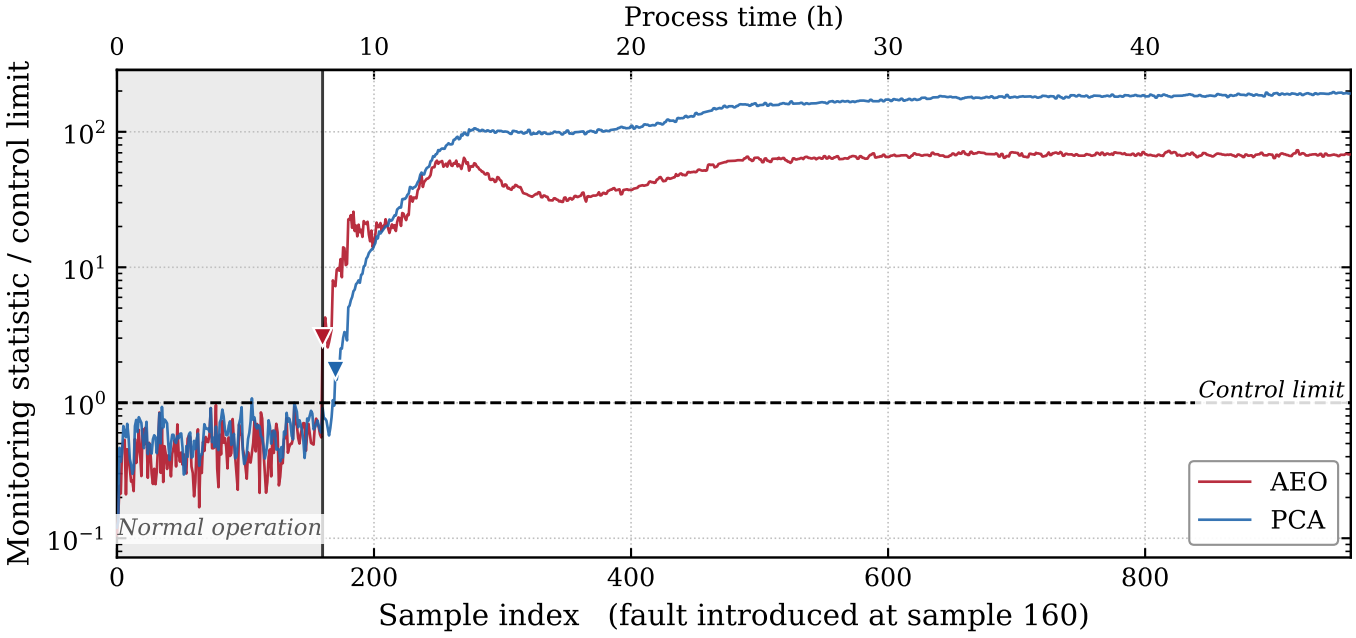
Fault Detection Performance for IDV(6)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	0.0 samples (0 min)	MDR 0.000	detected in 7/7 runs
PCA	delay	7.4 samples (22 min)	MDR 0.009	detected in 7/7 runs

AEO detects 7.4 samples (22 min) earlier; AEO missed-detection rate lower by 100%